1. "translation": $L : G \to \mathrm{Perm}(G)$ (an homomorphism) such that $L_g : x \mapsto g \cdot x$,[9]
2. "conjugation": such that $L_g : x \mapsto g \cdot x \cdot g^{-1}$.

6 Lesson 6

6.1 Special cases: boy!

Translations are always *transitive*: there is only an equivalence class (aka every element of G can be obtained from every (other) element of G by applying some translation).[1] For translations it is trivial to prove, because if

$$g \cdot x = y \implies g = y \cdot x^{-1}$$

because x always has an inverse.

Then, we can show that conjugation is a left action, because

$$L_{gg'} : x \mapsto (gg')c(gg')^{-1} = gg'x(g')^{-1}g^{-1} = g\big[g'x(g')^{-1}\big]g^{-1} = L_g(L_{g'}(x)) = L_g \circ L_{g'}(x),$$

thus it is an homomorphism.

Conjugation is not necessarily transitive: if G is an abelian group it is, for example, but in general it is not.

Translations applied to a proper subgroup $H \subset G$ are not transitive in general.

Definition 6.1 (Left coset)**.** Let $g \in G$, and consider the effect of translations L_g on the elements of a proper subgroup $H \subset G$. Then

$$gH = \{g' \in G : \exists H \in H, g' = L_g(h) = g \cdot h\}$$

is said to be a left coset.

In general, when H is a proper subgroup, there can be more than one left cosets. The set of all left cosets is called the *quotient space*

$$G/H = \{gH, \forall g \in G\}.$$

Proposition 6.2

The elements of each gH are in a one-to-one correspondence with the elements of H, and thus $|H| = |gH|$.

Proof. Consider $g \in G$ an define the mapping

$$\begin{aligned} f_g :& H \to gH \\ & h \mapsto g \cdot h. \end{aligned}$$

f_g is both injective

$$f_g(h_1) = f_g(h_2) \implies g \cdot h_1 = g \cdot h_2 \implies g^{-1} \cdot g \cdot h_1 = g^{-1} \cdot g \cdot h_2 \implies h_1 = h_2,$$

and surjective (by definition). This means that f_g is invertible, and forms a one-to-one correspondence between H and gH. □

[9] The nature of the definition is clear if we think about $G = \mathbb{R}^2$.

[1] Equivalently, $\forall x \in G, \forall y \in G, \exists g \in G : L_g(x) = y$.

Theorem 6.3 (Lagrange)
The order $|H|$ of any subgroup H of a finite group G must be a divisor of $|G|$.

Remark 6.4. Left cosets are the equivalence classes under the relation in G defined as: g and g' are equivalent if $g' = g \cdot h$ for some $h \in H$. Since equivalence classes are disjoint and their union reconstructs the entire set the sum of their elements (under the assumption that G, and thusly H, is finite), the sum of the orders of the gHs is equal to the order of G. However we learned that $|gH| = |g'H|, \forall g, g' \in G$.

Proof. Since G is the union of all gH, all left cosets are disjoint, and all have the same order

$$|G| = n|H|$$

with $n \in \mathbb{N}_0$. □

Remark 6.5 (Finite groups of prime order)**.** Does Z_5 have any proper subgroup? No, it is not possible: if it had another proper subgroup, it should have order greater than 1 and lower than 5, but 5 has no divisors that satisfy this property.

For a group with prime order, either $|H| = 1$ or $|H| = |G|$, but both of these facts are incompatible with the definition of proper subgroup.

Corollary 6.6
If p is prime, then $\mathbb{Z}_p$ is the only (up to isomorphisms) group with order p.

Proof. Assume that G is a group with order $|G| = p$, and let $g \in G$, with $g \neq e$ (because $p > 1$, if p is prime).

Consider the positive powers of g: g, g^2, g^3, ...there can only be a finite number $m - 1$ of powers that are different from the identity (and lower powers).[2] Thus, we can suppose that $g^m = e$.

We denote the following set

$$H = \{e, g, \dots, g^{m-1}\},$$

which is certainly a subset of G. If we take $h_1, h_2 \in H$, we have that $h_1 \cdot h_2 \in H$: they are all powers defined modulo m. Also, for all $g^k \in H$ we can define its inverse as $g^{m-k} \in H$. Therefore H is also a subgroup of G.

Finally, H was constructed isomorphically to $\mathbb{Z}_m$: the cyclic groups are precisely defined in this way. Since H is a subgroup of G, m must be a divisor of p: p is prime, and $m > 1$ (because we took $g \neq e$), therefore $p = m$.

Every element of G is contained in H, and H is isomorphic to $\mathbb{Z}_p$, thus G is isomorphic to $\mathbb{Z}_p$. □

Theorem 6.7
If X is a set and G is a group, such that it acts transitively on X, then G has a subgroup H such that G/H is in one-to-one correspondence with X.

[2] If there are finite elements, then it must be that $g^p = g^q$ for some $p > q$ (without loss of generality), thus $g^{p-q} = e$. If the order were not finite, this would not hold: $(\mathbb{Z}, +)$ is an example, because $1 + 1 + 1 + 1 + \dots$ does not give 0 eventually.

Consider an element $x \in X$, then by definition its orbit is

$$O_x = \{y \in X : \exists g \in G : y = L_g(x)\},$$

and the isotropy group

$$G_x = \{g \in G : L_g(x) = x\},$$

and thus G/G_x can be put in one-to-one correspondence with O_x.

In particular, two interesting results hold

1. the *orbit stabilizer theorem*: $|G| = |O_x| \cdot |G_x|$,
2. *Burnside's lemma*:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

with $X^g = \{x \in X : L_g(x) = x\}$.

If G is a group, and $H \subset G$ a proper subgroup, we defined gH and G/H: is G/H a group? In general, this is not the case, but it is true when H is a normal subgroup.

We can show this by defining the group multiplication in G/H as follows

$$(g_1 H) \cdot (g_2 H) = g_1 g_2 H,$$

which is clearly an internal operation, but luckily it satisfies also the other properties of the group product. It is not immediately clear, in part because a left coset is not precisely identified by an element g, because if

$$gH = \{ge, gh_1, gh_2, \ldots\}$$

then $gh_1 H = gH$: there is an ambiguity. It turns out that the condition to resolve this ambiguity is precisely the normality of H.

Suppose that we follow our definition

$$(g_1 H)(g_2 H) = g_1 g_2 H$$

and that

$$(g_1 h_1 H)(g_2 H) = g_1 h_1 g_2 H,$$

then

$$g_1 g_2 H \stackrel{?}{=} g_1 h_1 g_2 H$$

is true, since the normality of H implies that for some $h' \in H$

$$g \cdot h = h' \cdot g \implies g_1 h_1 g_2 = g_1 g_2 h_1'$$

for some h_1', and thus

$$g_1 h_1 g_2 H = g_1 g_2 h_1' H = g_1 g_2 H$$

using the ambiguity in our favour (because h_1' is multiplied on the right).

From this it follows that

$$|G/H| = \frac{|G|}{|H|}.$$

True? Is this equivalent to Lagrange? I don't think so

As an example we can consider

$$\mathbb{Z}_{2N} = \left\{e, a, a^2, \ldots, a^{2n-2}, a^{2n-1}\right\}$$

G	$Z(G)$
abelian group	G
$S_3 \cong D_3$	Z_1
D_4	$\mathbb{Z}_2$
$SO(2k+1, \mathbb{R})$	$\mathbb{Z}_1$
$SO(2k, \mathbb{R})$	$\mathbb{Z}_2$
$U(N)$	$U(1)$
$SU(N)$	$\mathbb{Z}_N$
$Sp(2N, \mathbb{R})$	$\mathbb{Z}_2$

Table 1: A few examples of groups and their centres. The other exotic example are $\mathbb{G}_2$, $\mathbb{F}_4$, $\mathbb{E}_6$, $\mathbb{E}_7$, $\mathbb{E}_8$, which respectively have $\mathbb{Z}_1$, $\mathbb{Z}_1$, $\mathbb{Z}_3$, $\mathbb{Z}_2$, $\mathbb{Z}_1$.

and its subgroup

$$H = \left\{e, a^2, a^4, \ldots, a^{2n-2}\right\},$$

which are both abelian, and consequently normal. We can construct the left cosets

$$eH (= a^2 H = a^4 H = \ldots) = H = \left\{e, a^2, \ldots, a^{2n-2}\right\}$$

and

$$aH (= a^3 H = a^5 H = \ldots) = \left\{a, a^3, \ldots, a^{2n-1}\right\},$$

while the quotient space

$$\mathbb{Z}_{2n}/H = \{eH, aH\}$$

is in fact a group (isomorphic to $\mathbb{Z}_2$), thankfully.[3]

Another interesting example is $G = SU(2)$ and $H = \{\mathbb{I}_2, -\mathbb{I}_2\}$ (where $H \subset G$, clearly). We note that $\forall A \in G$

$$A \cdot (\pm I_2) = \pm A = \pm \mathbb{I}_2 \cdot A,$$

and thus $\forall A \in G$, and $\forall h \in H$

$$A \cdot h \cdot A^{-1} = h \in H.$$

Since H is normal, $SU(2)/H$ is a group, and given that $H \cong \mathbb{Z}_2$

$$SU(2)/\mathbb{Z}_2 \cong SO(3)$$

which is a realisation of the facts we introduced about $SU(2)$ and $SO(3)$. Another interesting thing worth to mention: H contains all elements of G that commute with all $g \in G$. A(n abelian) subgroup with this property is called the *centre* of the group.

[3] It is important to note that a^2 can be different from e, but $a^2 H = eH$.